

Class 10 Science - Chapter 10

The Human Eye and the Colourful World (Hinglish Notes)

PART A - CORE BASICS (MISS MAT KARNA)

Ch9 + Class 8 ki yeh cheezein pehle pakki karo, warna Ch10 samajh nahi aayega.

1. Refraction aur Lens (Ch9 revision - bahut zaroori)

Refraction = light ka ek medium se doosre me mud-na (speed change).

Convex lens = beech mota, CONVERGING (kiranein milati) -> real image bana sakta.

Concave lens = beech patla, DIVERGING (kiranein failati) -> virtual chhota.

Power $P = 1/f$ (metre); unit DIOPTR (D). Convex +ve, concave -ve.

2. Image banna (Ch9 revision)

Lens screen par real, inverted image banata (jaise camera/aankh).

Focal length jitni chhoti -> power utni jyada (zyada mud-aata).

3. White light aur Colours (Class 8 base)

White light (suraj ki roshni) = saat (7) colours se bani.

VIBGYOR: Violet, Indigo, Blue, Green, Yellow, Orange, Red.

Prism = teen-konaa (triangular) kaanch jo light ko todta.

Transparent medium me alag colour alag speed se chalte.

4. Atmosphere aur kann (Class 7-8)

Atmosphere = Earth ke charo taraf hawa (gases + dust + paani ke kann).

Light chhote kanno se takrakar bikhar (scatter) sakti.

Chhoti wavelength (blue) zyada bikharti, badi (red) kam bikharti.

5. Wavelength ka idea

Light ek wave hai; har colour ki alag wavelength.

Red ki sabse BADI wavelength, Violet ki sabse CHHOTI.

Chhoti wavelength = zyada scatter (bikhrav).

-- END OF CORE BASICS (RED) PAGE --

PART B - CHAPTER KA MAIN CONTENT

Class 10 Science | Ch 10: The Human Eye and the Colourful World

[GREEN = board exam me baar-baar poocha jaata hai]

1. Human Eye ki Structure [VERY IMPORTANT - RATTA]

CORNEA: aankh ke aage ka transparent parda; sabse zyada refraction yahin.

IRIS: rangeen hissa; pupil ka size control karta.

PUPIL: beech ka kaala chhed; kitni light andar aaye control (iris se).

EYE LENS (crystalline): convex lens; fine focusing karta (image sharp).

CILIARY MUSCLE: lens ki shape (focal length) badalta.

RETINA: peeche ka parda (screen) jahan REAL, INVERTED image banti.

Isme light-sensitive cells: RODS (kam roshni/dim) + CONES (rang + tej roshni).

OPTIC NERVE: image ka signal brain tak le jaata.

Aankh ki image: retina par real aur inverted hoti (brain seedha samajhta).

2. Power of Accommodation [IMPORTANT]

Accommodation = aankh ke lens ka apni FOCAL LENGTH badal-ne ki capacity, taaki paas aur door dono cheez sharp dikhe.

- Door dekhne par: ciliary muscle DHEELE -> lens patla -> focal length BADI.

- Paas dekhne par: ciliary muscle TIGHT -> lens mota -> focal length CHHOTI.

Near point (least distance of distinct vision) = 25 cm (normal aankh).

Far point = infinity (normal aankh door tak dekh sakti).

Lens itna patla nahi ho sakta ki 25cm se paas focus kare -> isliye bahut paas ki cheez clear nahi dikhti.

3. Defects of Vision aur Sudhaar (Correction) [TOP EXAM AREA]

(A) MYOPIA (nikat-drishti / near-sightedness):

- PAAS saaf dikhta, DOOR nahi.

- Kaaran: image retina ke AAGE banti (eyeball lamba ya lens zyada convex).

- Sudhaar: CONCAVE lens (diverging) - far point wapas infinity par.

(B) HYPERMETROPIA (door-drishti / far-sightedness):

- DOOR saaf dikhta, PAAS nahi.

- Kaaran: image retina ke PEECHE banti (eyeball chhota ya lens kam convex).

- Sudhaar: CONVEX lens (converging) - near point 25cm par le aata.

(C) PRESBYOPIA:

- Budhape me aankh ke muscle kamzor + lens lachila nahi -> paas+door dono me dikkat.

- Sudhaar: BIFOCAL lens (upar concave for door, neeche convex for paas).

Cataract: lens dhundhla (cloudy) ho jaata -> surgery se theek (yeh accommodation defect nahi, alag).

Memory: "Myopia = Minus (concave) lens; Hypermetropia = + (convex) lens".

4. Refraction through Prism aur Dispersion [TOP EXAM AREA]

Prism light ko mod-ta (do baar refraction) aur DISPERSION karta:

DISPERSION = white light ka apne 7 colours (VIBGYOR) me toot jaana.

Kaaran: har colour alag angle se mudta (Violet sabse zyada, Red sabse kam).

Spectrum = VIBGYOR ki rangeen patti.

Recombination: doosra ulta prism lagao -> 7 colour wapas white light ban jaate (Newton ka experiment).

5. Rainbow (Indradhanush) [IMPORTANT]

Rainbow = baarish ke baad aakash me VIBGYOR ka arc.

Kaise banta: hawa me paani ki boondein chhote PRISM jaise kaam karti ->

sunlight me REFRACTION + DISPERSION + INTERNAL REFLECTION + REFRACTION ->
7 colour alag dikhte. Suraj peeche, boondein saamne hone par dikhta.

6. Atmospheric Refraction [IMPORTANT]

Atmosphere me hawa ki layers ka temperature/density alag -> light mud-ti.

(A) Tare timtimate (twinkling) - star ki light layers se mud-mud ke aati, thodi-thodi badalti -> kabhi tej kabhi mand. (Planet nahi timtimate kyunki woss paas hain -> bade source.)

(B) Suraj sunrise/sunset par jaldi dikhna - actual horizon ke neeche hone par bhi atmospheric refraction se 2 min pehle dikh jaata.

(C) Sunrise/sunset par suraj OVAL/laal dikhna.

7. Scattering of Light (Tyndall + Sky colour) [TOP EXAM - VERY IMP]

Scattering = light ka chhote kanno se takrakar har disha me bikhar jaana.

TYNDALL EFFECT = colloid/dust me light ka path dikhna (jaise jungle me dhundh me sunlight ki kiran, ya smoke me torch beam).

Chhoti wavelength (BLUE) zyada scatter; badi (RED) kam scatter.

(A) Aakash NEELA kyun? - hawa ke molecules blue (chhoti wavelength) ko zyada scatter karte -> aankh tak chaaro taraf se blue aata -> sky blue dikhta.

(B) Sunrise/sunset par suraj LAAL kyun? - tab light ko zyada hawa paar karni padti -> blue raaste me hi scatter ho jaata -> sirf RED (kam scatter) bachta.

(C) Antariksh (space) me aasman KAALA - kyunki wahan hawa/molecules nahi -> scattering nahi.

(D) Khatre ka signal LAAL kyun? - red light kam scatter hoti -> door se bhi saaf dikhti -> isliye danger sign red.

8. Quick Revision - One Liners

- Eye: cornea (max refraction), iris (pupil control), retina (real inverted image, rods+cones), accommodation = lens focal length change.
- Near point 25cm, far point infinity.
- Myopia -> concave lens; Hypermetropia -> convex; Presbyopia -> bifocal.
- Dispersion: white -> VIBGYOR (violet max bend, red min).
- Sky blue + sunset red = scattering (blue zyada, red kam scatter).

PART C - SOLVED EXAMPLES (HARDEST -> EASIEST)

Example 1 (HARDEST) - Defect identify + lens + kaaran

Q: Ek vyakti door ki cheez saaf nahi dekh paata par paas ki dekh leta. Kaunsa defect? Kaaran kya? Kaunsa lens? Image kahan banti?

Solution:

- Defect: MYOPIA (near-sightedness).
- Kaaran: eyeball lamba ya lens zyada convex -> image retina ke AAGE banti.
- Sudhaar: CONCAVE (diverging) lens - jo door ki kiranein thodi failakar image wapas retina par laata.

Example 2 - Sky blue + sunset red dono ek saath

Q: Aakash din me neela par sunset par suraj laal kyun dikhta? Scattering se samjhao.

Solution:

- Blue ki wavelength chhoti -> hawa molecules zyada scatter karte -> din me chaaro taraf se blue aata -> sky blue.
- Sunset par light ko zyada atmosphere paar karna padta -> blue raaste me scatter ho jaata, sirf RED (kam scatter) aankh tak pahunchta -> suraj laal.

Example 3 - Dispersion + recombination

Q: Prism se white light ka kya hota? Wapas white kaise milegi?

Solution:

- White light prism se guzar kar 7 colours (VIBGYOR) me TOOT jaati (dispersion) - violet sabse zyada, red sabse kam mudta.
- Ek doosra ulta (inverted) prism lagao -> saare colour milkar wapas WHITE light ban jaati (Newton).

Example 4 - Accommodation

Q: Power of accommodation kya hai? Door aur paas dekhte waqt lens me kya change?

Solution:

- Accommodation = lens ka focal length badalne ki capacity.
- Door: ciliary muscle dheele -> lens patla -> focal length badi.
- Paas: ciliary muscle tight -> lens mota -> focal length chhoti.

Example 5 (EASIEST) - Retina par image

Q: Aankh me image kahan banti aur kaisi (real/virtual)?

Solution:

- RETINA par; real aur inverted (ulti).

PART D - SELF TEST (EASY -> HARD)

EASY (1 mark)

- Q1. Retina ka kaam kya hai? Rods aur cones me antar.
- Q2. Normal aankh ka near point aur far point.
- Q3. Tyndall effect kya hai? 1 example.
- Q4. VIBGYOR me konsa colour sabse zyada/kam mudta.

MEDIUM (2-3 marks)

- Q5. Power of accommodation samjhao (door vs paas).
- Q6. Myopia ka kaaran + sudhaar (lens).
- Q7. Hypermetropia ka kaaran + sudhaar (lens).
- Q8. Dispersion kya hai? Prism se kaise hoti.

HARD (3-5 marks, board favourite)

- Q9. Aankh ki structure (5 bhaag + kaam).
- Q10. 3 vision defects (myopia/hypermetropia/presbyopia) + lens.
- Q11. Aakash neela + sunset laal scattering se explain.
- Q12. Atmospheric refraction: tare timtimana + suraj jaldi dikhna.

ANSWER HINTS (PART D)

- A1. Retina = screen (real inverted image); rods (dim light), cones (rang).
- A2. Near point 25 cm, far point infinity.
- A3. Colloid/dust me light path dikhna; jungle dhundh me sunlight kiran.
- A4. Violet sabse zyada, Red sabse kam.
- A5. Lens focal length change; door-patla(badi f), paas-mota(chhoti f).
- A6. Image retina ke aage; concave lens.
- A7. Image retina ke peeche; convex lens.
- A8. White light ka 7 colour me toot-na; prism alag-alag angle se modta.
- A9. Cornea, iris, pupil, lens, retina, optic nerve (+kaam).
- A10. Myopia-concave; hypermetropia-convex; presbyopia-bifocal.
- A11. Blue zyada scatter (sky blue); sunset blue scatter ho jaata, red bachta.
- A12. Layers se light mudti -> tare timtimana; suraj 2 min pehle dikhta.

"Padhai tabhi pakki jab khud likh ke test do." - All the best, Krishna!

Generated by Kiro for Krishna1k
